

Next-Generation Public Wi-Fi Security using SDRN-based Early Attack Detection

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Abstract— The rise in popularity of public Wi-Fi networks has led to a significant increase in the number of exposure points to inchoate attacks, such as rogue access points, evil twins, and ARP spoofing, which present a significant threat to users of public Wi-Fi due to the potential of utilizing a man-in-the-middle or denial-of-service attack on unsuspecting users. Traditional methods of protecting against these types of attacks utilize single-layer based monitoring systems or signature-based detection methodologies, which are slow to respond to intrusions and are inconsistent with the current ever-changing nature of most public Wi-Fi networks. This paper presents a next-generation public Wi-Fi security framework utilizing a Software-Defined Radio Network (SDRN) platform for the purpose of early detection of inchoate attacks. The architecture that the proposed system employs is based on an integration of physical layer beacon frame fingerprinting, link layer anomaly analysis, and network layer ARP behaviour monitoring to detect malicious activity at the onset. Through the independent analysis of cross-layer features and fusion of these features at the decision level, the proposed system improves the accuracy of detection while reducing false positive results. With the centralized administration and real-time reconfigurability capabilities of SDRN, the proposed approach facilitates the rapid isolation of rogue access points and the mitigation of spoofing attacks. The experimental evaluation of the proposed framework has been shown to significantly improve both the latency and robustness of detection compared with traditional Wi-Fi intrusion detection approaches and provides appropriate protection for large scale public wireless environments.

Keywords— *Software-Defined Radio Networks (SDRN), Wi-Fi Security, Rogue Access Point Detection, Beacon Frame Fingerprinting, ARP Spoofing Detection, Cross-Layer Intrusion Detection, Decision-Level Fusion*

I. INTRODUCTION

Public Wi-Fi networks are commonly deployed in open spaces and are very vulnerable to security threats that include rogue access points, evil twin attacks, and ARP spoofing. Attacks such as these allow man-in-the-middle interception of traffic, manipulation of traffic (e.g., from one system to another), and denial-of-service (DOS) attacks; this is often done without detection by traditional security products. For example, it has been shown that traditional closure products

(e.g., firewalls) do not offer much in terms of detecting attacks.

The existing solutions for detecting Wi-Fi intrusions typically rely on a single layer of protection (one method of monitoring) or signature-based detection (e.g., identifying a signature in a traffic event), which results in very limited visibility/monitoring capability for detecting attacks; delayed detection of attacks; and very low adaptability to the rapidly changing nature of wireless networks. Moreover, as attackers become more sophisticated and exploit vulnerabilities across multiple layers of the OSI model, isolated methods of detection will become ineffective.

Software-Defined Radio Networks (SDRNs) provide a means of providing centralized control and cross-layer visibility; therefore, they enable real-time monitoring of the physical, link, and network layers. Using the advantages of SDRN technology, this performance paper describes how to develop a central attack-detection framework for public Wi-Fi networks. In this framework, we propose the early detection of rogue access points (RAPs) and ARP attacks by using beacon frame fingerprinting and ARP behavior analysis, and we fuse decision-making mechanisms to provide accurate early detection of both RAPs and ARP spoofing attacks. By conducting experiments to test the developed framework (compared to traditional detection methods), we demonstrate that the SDRN attack detection framework can provide significantly better accuracy and significantly lower detection time than detection methods typically employed in public Wi-Fi networks.

II. RELATED WORKS

There has been a great deal of research done into wireless network vulnerabilities, as well as new methods to detect/mitigate attacks including but not limited to ARP spoofing, rogue access points (RAP), and evil twins (ET). The two most common examples of how ARP is exploited to conduct MITM attacks are through ARP spoofing and ARP poisoning. To this end, Chandru and Bagyalakshmi [1] created a detection/prevention framework aimed at MITM attacks via ARP poisoning in wireless networks by focusing on the need to identify abnormal ARP behavior. Likewise, J.P.S. and Sudha [2] propose ARP spoofing detection mechanisms in

order to prevent DDoS attacks caused by ARP manipulations. Alternately, Patel and Shah [3] performed a comparative analysis of ARP spoofing detection methods, pointing out the shortcomings of traditional static threshold methods in dynamic networking environments. A separate detailed survey of ARP spoofing detection/mitigation strategies provides evidence of the advancement of ARP security solutions [9].

There has been a lot of research on the topic of unauthorized access points (APs) and evil twin attacks in IEEE 802.11 wireless networks. Yang et al. (2015) use deep learning, purging physical-layer features, and knowing the device's identity to eliminate rogue AP attacks in Internet of Things (IoT) settings. Igarashi et al. (2011) provide a method for identifying rogue APs based on the use of ARP failures due to MAC address duplication, as well as demonstrating that protocol-level anomalies can be used to authenticate users using lower level protocols. Komanduri and Sankaran (2013) conduct experiments on the vulnerabilities of WLANs to rogue APs and highlight the practical concerns for their deployment. Jang et al. (2015) extend this work by exposing rogue APs using deliberate interference with their communication channels and thus provide a physical layer perspective. Selvarathinam et al. (2013) model evil-twin detection using discrete event systems and Esser and Serrao (2014) develop an integrated framework for testing evil twins to evaluate the performance of detection systems.

Wireless LAN threat landscapes have been contextualised through comprehensive reviews. Vaidya et al. (13), examined and categorised spoofing detection methods used with wireless LANs and identified research gaps. Conti et al. (14), have conducted a survey of man-in-the-middle attacks within wireless systems and provided an outline of the key attack vectors and defensive strategies. Alotaibi and Elleithy (15), have examined empirical fingerprinting frameworks for detecting rogue access points (APs) and provided early work on device-level authentication. Survey articles on the security of Software-defined Networks (SDN) further established the foundation for methods to control programmable networks.

ARP Protocol's Security Measures include Secured ARP Protocol Designs and SDN (Software Defined Network) enabled defenses. AbdelSalam and others (2005) have proposed methods to help mitigate the risk associated with ARP Spoofing in Software Defined Networks using centralized control of ARP and the centralized enforcement of proper Address Resolution Protocol (ARP) usage. Ibrahim et al. (2007) present an SDN (software defined network) approach for limiting the impact of ARP Spoofing attacks through the use of Centralized ARP data and controlling the Flow of broadcast traffic in the form of "Flooding of Network Traffic". There are classical (traditional) ARP protocols such as S-ARP that have established a foundation for the security of all Address Resolution Protocol (ARP) solutions (Scott-Hayward et al., 2008). The research obtained from Scott-Hayward and others (2008) identified that security in an SDN environment has many challenges associated with it and must be considered if developing software programmed defenses to Wireless Networks.

To improve the security of wireless networks, several techniques have been developed that use physical layer and device fingerprinting as additional methods to improve wireless security. Shi & Jensen (2014) have developed a method to improve radiometric identification by using MIMO

transmissions, thereby extending the application of physical fingerprinting to multiple antenna systems. Jana & Kasera (2005) have used clock skew characteristics of APs to perform unauthorized AP detection, which indicates that low-level signal characteristics can be used to verify authenticity. Sheng et al. (2008) have used patterns in received signal strength indicator (RSSI) values to detect MAC spoofing, which provides an association between physical measures and the security outcomes at higher protocol layers. Rasmussen & Capkun (2006) have explored the use of radio fingerprinting for the security of sensor networks, thereby providing motivation for multi-layer security analysis.

Studies have been done on intrusion detection systems for wireless networks, which provide frameworks for misuse and anomaly detection. For instance, the authors Raya et al. [23] have examined the behavior of greedy MAC protocols at hotspot locations, whereas Kyasanur and Vaidya [25] have focused on MAC layer misbehavior. Both works contributed to understanding of the security threats faced by link-layers. Additionally, many security monitoring mechanisms provide a much larger context through sensor network security surveys [20] as well as details of the intrusion detection process [31] including anomaly detection techniques [33]. The design of a secure ARP protocol and detection of ARP spoofing attacks [24, 22 respectively] represents examples of protocol-based defense strategies. The detection of unauthorized wireless access points via RSSI [21] along with sensor network security surveys [20] has provided additional support for the importance of utilizing cross-layer analysis as a method of achieving secure wireless networks.

Although there has been a good deal of progress in all these areas, most of the existing solutions are based on single-layer analysis, or use constant thresholds (rather than adaptive ones), or depend on a machine learning model that may not be effective across many different types of public Wi-Fi environments. The proposed approach will combine two different kinds of cross-layer fusion to create a more accurate and resilient detection technique when used in an untrusted and/or dynamic public Wi-Fi environment. By using physical layer beacon fingerprinting at the physical layer (Layer 1), and network layer ARP analysis at the network layer (Layer 3), the new approach will provide enhanced detection and more robust performance.

There are also significant limitations in the existing body of knowledge due to a lack of an integrated decision - level analysis across multiple protocol levels and a correlation of data in real - time from all of these levels. The majority of the studies utilize only isolated performance metrics or offline test data to validate their methodologies and provide little insight into the temporal behavior of attacks, detection latency, and false alarm characteristics given realistic traffic flows. Similarly, machine - learning approaches typically require very large datasets, significant computational requirements or special hardware, making it unlikely that these would be able to support lightweight monitoring applications such as public Wi - Fi hotspots. Visualizations and feedback to operators are also frequently neglected even though they are essential for the practical deployment of any system and timely incident response. The proposed system seeks to fill these gaps by providing a passive, low - complexity, non - intrusive monitoring approach based on cross - layer feature extraction, decision level fusion and intuitive visualization to allow

accurate and timely detection of rogue APs and ARP - based man - in - the - middle (MITM) attacks in real - world wireless environments.

III. PROPOSED SYSTEM

A. Overview of the Proposed System

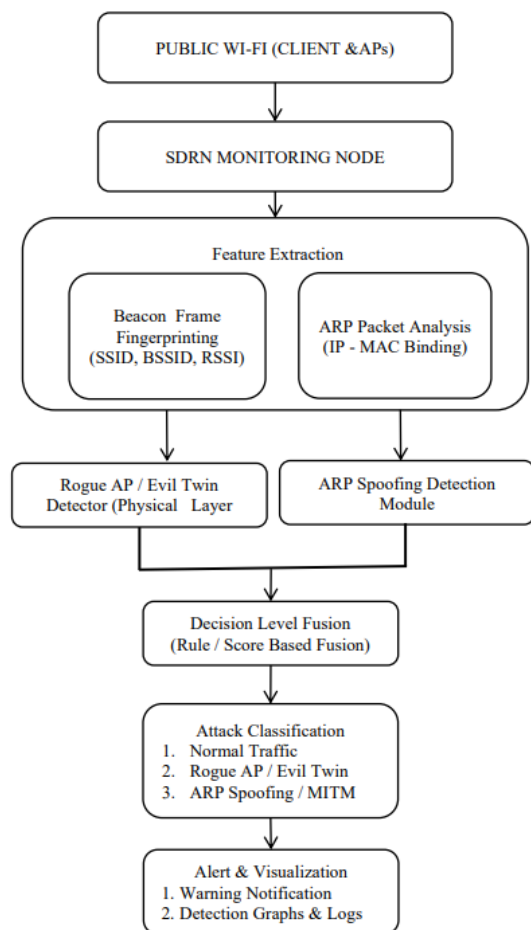


Figure 1. Systematic architecture of the proposed SDRN-based public Wi-Fi attack detection framework.

The suggested new system provides an early warning framework for the detection of attacks via attack detection systems that are used with Software Defined Radio Networks (SDRN) (i.e., Access Points, Evil Twin Access Point, and ARP Spoofing) by helping to secure public Wi-Fi networks against different types of attacks. The proposed system performs monitoring of wireless traffic in both the physical and data link layers without interfering with user to user communications. It performs independent analysis of the physical layer characteristics of beacons for each type of wireless access point and the ARP packet characteristics of the received packets at the data link layer for each of the wireless access points being evaluated. The combination of both the results of the independent analyses of the two detection systems provides a means of increasing detection reliability and reducing false positives through the use of a decision level fusion technique. The combined output of both detection channels produces a classification for the network state and provides real-time alerts and visual analytic representations to the network administrator.

Figure 1 illustrates the end-to-end architecture of the proposed system, as shown by the diagram for traffic monitoring through the attack classification and alert generation stage of the system.

B. Architecture Description

1) *Public Wi-Fi Environment*: The public Wi-Fi environment consists of legitimate access points, client devices, and other potential attacker nodes all within the same wireless domain. Public Wi-Fi has a great deal of vulnerability to rogue access point deployments and ARP spoofing attacks due to its open and unregulated status, which makes it the ideal environment to evaluate our detection system.

2) *SDRN Monitoring Node*: SDRN monitoring nodes utilize software defined radio (SDR) techniques to passively monitor and capture real-time wireless and network-layer traffic. Non-intrusive monitoring occurs as only beacon frames and ARP packets are captured – this process does not interfere with the normal operations of the network.

3) *Features Extraction Module*: After we capture the traffic, this module will analyze and extract attack-relevant features by employing parallel physical layer and network layer analysis of the captured data.

a) *Beacon Frame Fingerprinting*: The fingerprinting of the beacon frames is accomplished by utilizing the SSID, BSSID, and RSSI parameters of the beacon frame. Therefore, the fingerprints of legitimate access points should be stable while the fingerprints of a rogue or evil twin access point will exhibit significant variations. Thus, by using stable fingerprints, we can identify physical-layer attacks.

b) *ARP Packet Analysis*: The ARP packets captured during the monitoring process can be inspected to detect inconsistency in the binding between an IP and MAC address. If a MAC address binds to an IP address that is also bound to other MAC addresses, this would imply the possibility of ARP spoofing or MITM activity occurring.

4) *Detection Modules*: The two detection modules function independently and in parallel. The rogue access point detection module uses access point (AP) beacon fingerprints to identify wireless APs that are unauthorized. The ARP spoofing detection module detects attacks occurring on the network layer through binding violations.

5) *Decision Level Fusion*: The physical and network layers' detection output is combined in the decision-level fusion block using score or rule-based fusion logic. This process helps improve detection accuracy through correlation of the evidence of both layers, reduces false alarms and provides a higher level of robustness when it comes to classifying multiple types of attacks (e.g., DoS, etc.) within an environment with high levels of noise or a dynamic network.

6) *Attack Classification*: The fused decision enables the system to determine the state of the network by classifying it as one of only a few different states:

- Normal Traffic
- Rogue AP/Evil Twin Attack
- ARP Spoofing/MITM Attack

This classification provides accurate identification of both the attack type and operational layer associated with it.

7) *Alert and Visualization Module*: The last block produces warnings to administrate, it also provides visual analytic data to help them with their situation awareness through detections timelines, anomaly scores and logs for forensic capabilities. This module will assist with responding quickly and eliminating any threats.

C. Operational flow

Figure 2 shows the workflow of an SDRN-based Wi-Fi attack detection system. The first event in this process is the constant observation of a public Wi-Fi environment: legitimate access points and client devices connecting and transferring information in an open wireless environment. An SDRN monitoring node captures wireless beacon frame transmissions from all access points, as well as wirelessly transmitted ARP packet frames from all devices on the network, without disrupting normal operation on the network.

Following capturing, data are sent to the feature extraction stage, at which point the captured data undergo two parallel processes. In the first process, wireless beacon fingerprints are extracted by determining the access point's (AP's) physical layer features (such as SSID, BSSID, and RSSI) in order to identify abnormal access point activity. The other parallel process is to analyze ARP packets to identify discrepancies with the IP/MAC address bindings as a result of either spoofing attacks or man-in-the-middle attacks.

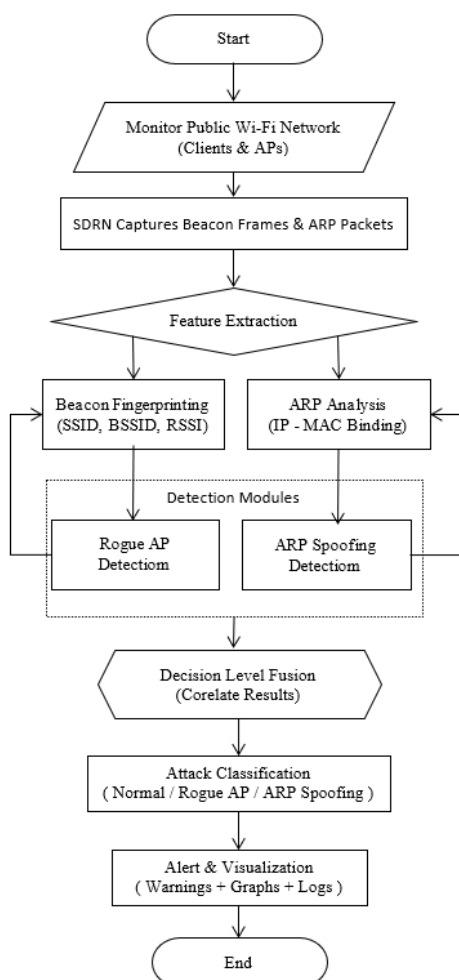


Figure 2. Workflow of the proposed SDRN-based early attack detection system for public Wi-Fi networks.

Detection modules are used to process the features that have been extracted from the different access points. The recognition of "evil twin" (rogue) access points occurs through the use of a beacon anomalous detection module. The ARP spoofing detection module is able to determine IP-MAC address conflicts using layer 2 of the protocol stack (network layer). The output from both the rogue AP and ARP spoofing detection modules will then be combined by the decision level fusion technique in order to provide a higher reliability rate and reduce the number of false alarms.

Based upon the combined decision, the overall network traffic will be classified into three categories: normal traffic, rogue AP/evil twin attack, or ARP spoofing. Finally, real-time monitoring and post-event analysis will be supported through the generation of alerts and visual analytics (i.e., warning messages, detection graphics, logs, etc.) as shown in the workflow diagram (Figure 2).

IV. RESULTS AND DISCUSSION

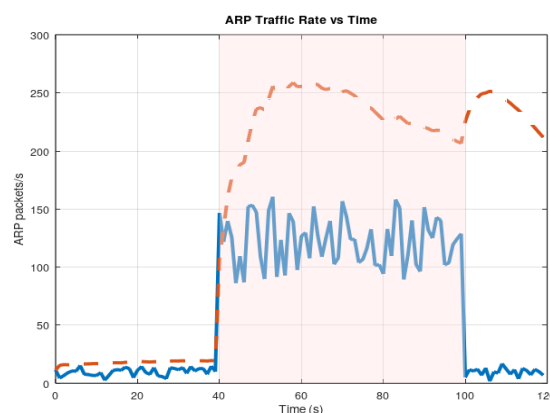


Figure 3. ARP traffic rate variation over time under normal and attack conditions, highlighting abnormal packet bursts during ARP spoofing.

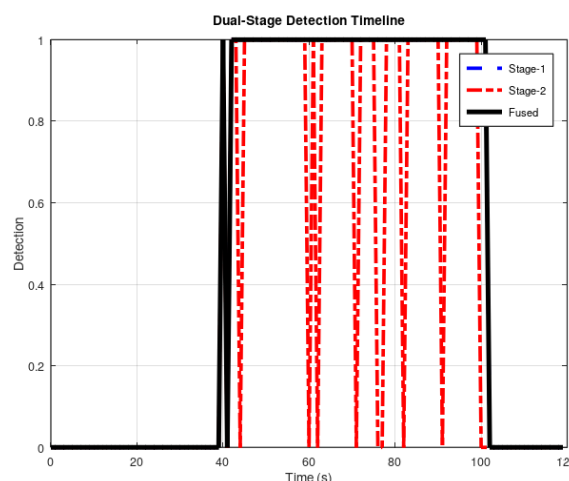


Figure 4. Detection timeline showing Stage-1 (beacon fingerprinting), Stage-2 (ARP analysis), and fused detection outputs.

Through numerous performance charts produced from the simulation results, we have validated the proposed SDRN two-stage detection strategy's effectiveness by analysing the attack behaviour of the system, its ability to respond, and its classification accuracy.

Fig. 1 displays the ARP traffic's changing rate over a period of time; at the point of normal operation, ARP packet rate is stable and consistently remains below predetermined threshold values. As soon as either an ARP spoofing or a Man-In-The-Middle attack is perpetrated, we see an immediate and significant increase in the rate of ARP traffic (ARP packet rate) which will directly lead to a corresponding number of threshold violations. This is evidence that abnormal patterns of ARP traffic can be used as strong indicators of network layer attacks.

The detection timeline shown in Figure 4 further details how the dual-staged detection process performed during an attack event. The physical layer detection (rogue APs via beacon fingerprint deviations) and the network layer (ARP inconsistencies) produce detection outputs that converge more rapidly with greater stability when they are combined than when they are processed separately; both are factors indicating the benefits of using decision-level fusion to reduce instances of false-positive results.

Figure 6 shows the ROC curve for the proposed system. This curve is really close to the left corner. This means the proposed system has a true positive rate and a low false positive rate for all thresholds. The proposed system is very good at telling the difference, between activity and legitimate traffic using the fused detection method. The fused detection method is an effective way to do this.

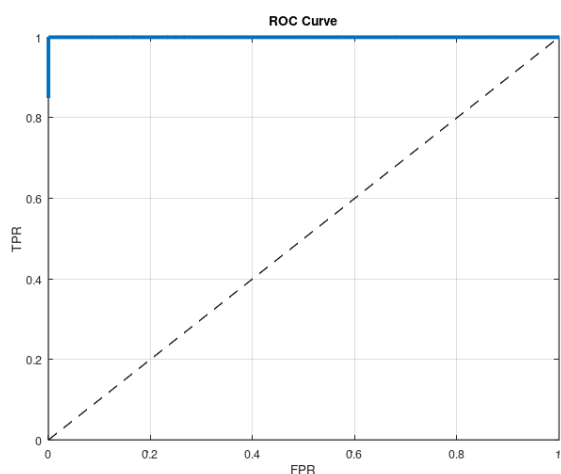


Figure 5. Receiver operating characteristic (ROC) curve of the proposed SDRN-based detection system.

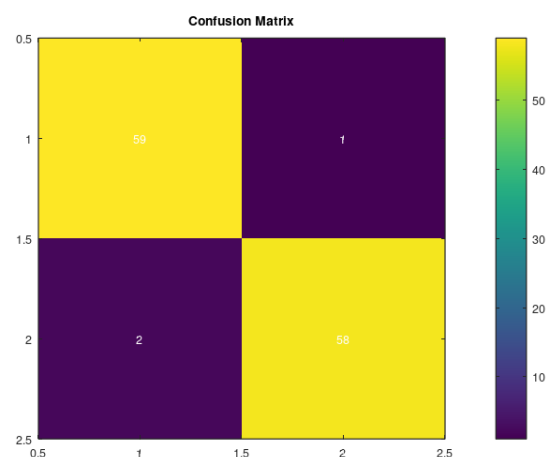


Figure 6. Confusion matrix of the fused detection model showing classification performance.

The classification performance of the system is summarized in the confusion matrix in Figure 7, where the large proportion of TPs and TNs confirms that attack and normal scenarios are correctly classified, while only a few instances of false positives and false negatives indicate the reliability and resiliency of the detection framework under different network states.

In summary, the analysis of physical layer beacons and monitoring of ARP over the network layer significantly improve the accuracy and resilience of the detection results, therefore providing a viable solution to defend public Wi-Fi networks against rogue APs and ARP spoofing attacks

V. CONCLUSION

The security system we developed can spot Wi-Fi access points and ARP spoofing attacks in public Wi-Fi networks. We used a software-defined networking approach to keep an eye on these activities. This approach combines information from layer beacons and network layer ARP to detect attacks without changing the network setup. Our simulation results show that weird ARP traffic patterns and inconsistent beacons can indicate activity. Using both methods to detect breaches gives us accurate results, fewer false alarms and better stability across different types of networks. The dual-layer security framework helps to identify access points and ARP spoofing attacks. The study utilizes a software defined networking based approach to monitor access points and ARP spoofing attacks combining physical layer beacons and network layer ARP information. The combination of both methods, for detecting breaches leads to an increased accuracy. The dual-layer security framework and software defined networking based approach provide detection of access points and ARP spoofing attacks. The security system provides stability amongst the varying nature of networks. Leads to less false alarms. Testing of the systems using detection timelines, ROC analysis, and confusion matrices reveal that the proposed framework is robust and effective in enhancing situational awareness and security in open wireless networks and is appropriate for deployment within public Wi-Fi environments. The proposed systems will be evaluated by future works that include implementation of hardware solutions and adaptive learning

mechanisms to improve detection performance within larger scale and more mobile environments.

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